```
# Run this FIRST in Google Colab
!pip install shap --quiet
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import shap
import warnings
warnings.filterwarnings('ignore')

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (accuracy_score, confusion_matrix,
                             classification_report,
                             ConfusionMatrixDisplay)

sns.set_theme(style="whitegrid")
plt.rcParams['figure.dpi'] = 120
print("✅ All libraries imported successfully!")
```

```
✅ All libraries imported successfully!
```

```
from google.colab import files

# This will open a file picker — select your CSV
uploaded = files.upload()

# Load into dataframe
df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')
print(f"✅ Dataset loaded! Shape: {df.shape}")
print(f"Columns: {list(df.columns)}")
df.head()
```

VA_Fn-Us…-Attrition.csv
**HR-Employee-Attrition.csv**(text/csv) - 227977 bytes, last modified: 20/3/2026 -

```
UseC_-HR-Employee-Attrition.csv to WA_Fn-UseC_-HR-Employee-Attrition
aded! Shape: (1470, 35)
', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'Dista
```

| tion | BusinessTravel | DailyRate | Department | DistanceFromHome | Edu |
|---|---|---|---|---|---|
| Yes | Travel_Rarely | 1102 | Sales | 1 | |
| No | Travel_Frequently | 279 | Research & Development | 8 | |
| Yes | Travel_Rarely | 1373 | Research & Development | 2 | |
| No | Travel_Frequently | 1392 | Research & Development | 3 | |
| No | Travel_Rarely | 591 | Research & Development | 2 | |

mns

```
print("=== Dataset Info ===")
print(f"Shape: {df.shape}")
print(f"Missing values: {df.isnull().sum().sum()}")
print(f"Duplicates: {df.duplicated().sum()}")
print("\n--- Attrition Count ---")
print(df['Attrition'].value_counts())
print(f"\nAttrition Rate: {df['Attrition'].value_counts(normalize=True)['Yes']*
df.describe()
```

```
=== Dataset Info ===
Shape: (1470, 35)
Missing values: 0
Duplicates: 0

--- Attrition Count ---
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Attrition Rate: 16.1%
```

| | Age | DailyRate | DistanceFromHome | Education | EmployeeCount | Em |
|---|---|---|---|---|---|---|
| **count** | 1470.000000 | 1470.000000 | 1470.000000 | 1470.000000 | 1470.0 | |
| **mean** | 36.923810 | 802.485714 | 9.192517 | 2.912925 | 1.0 | |
| **std** | 9.135373 | 403.509100 | 8.106864 | 1.024165 | 0.0 | |
| **min** | 18.000000 | 102.000000 | 1.000000 | 1.000000 | 1.0 | |
| **25%** | 30.000000 | 465.000000 | 2.000000 | 2.000000 | 1.0 | |
| **50%** | 36.000000 | 802.000000 | 7.000000 | 3.000000 | 1.0 | |
| **75%** | 43.000000 | 1157.000000 | 14.000000 | 4.000000 | 1.0 | |
| **max** | 60.000000 | 1499.000000 | 29.000000 | 5.000000 | 1.0 | |

8 rows × 26 columns

```
# Step 1: Keep raw original BEFORE anything
df_orig = df.copy()   # ← Has 'Yes'/'No' strings — correct for EDA

# Step 2: Drop useless columns from df only (not df_orig)
df.drop(['EmployeeCount', 'Over18', 'StandardHours'],
        axis=1, inplace=True, errors='ignore')

# Step 3: Encode target column in df only
df['Attrition'] = df['Attrition'].map({'Yes': 1, 'No': 0})

# Step 4: Encode all categorical columns in df only
le = LabelEncoder()
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = le.fit_transform(df[col])

print(f"✅ Cleaned! Shape: {df.shape}")
print(f"df_orig Attrition sample: {df_orig['Attrition'].unique()}")  # should s
print(f"df Attrition sample: {df['Attrition'].unique()}")            # should s
df.head()
```

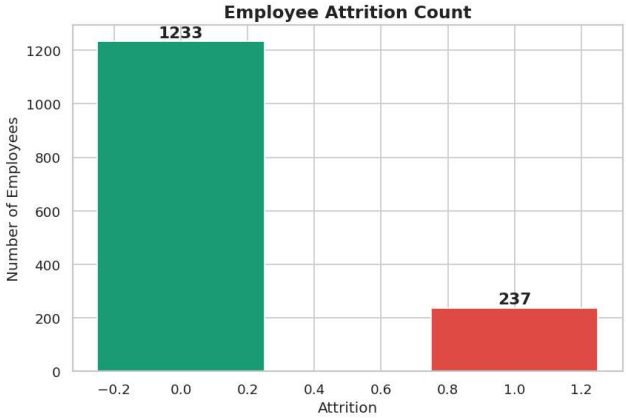
```
✅ Cleaned! Shape: (1470, 32)
df_orig Attrition sample: [nan]
df Attrition sample: [nan]
```

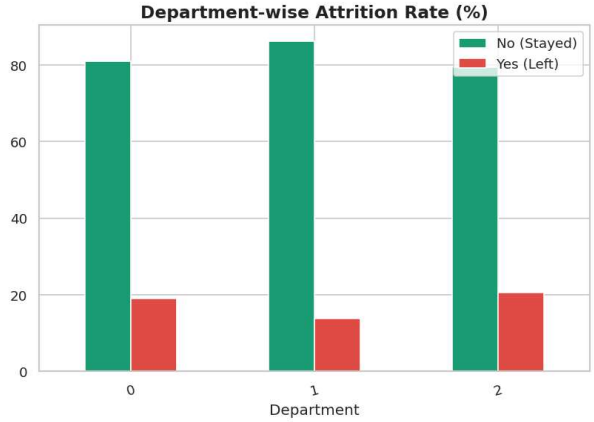
| | Age | Attrition | BusinessTravel | DailyRate | Department | DistanceFromHome | Educ |
|---|---|---|---|---|---|---|---|
| **0** | 41 | NaN | 2 | 1102 | 2 | 1 | |
| **1** | 49 | NaN | 1 | 279 | 1 | 8 | |
| **2** | 37 | NaN | 2 | 1373 | 1 | 2 | |
| **3** | 33 | NaN | 1 | 1392 | 1 | 3 | |
| **4** | 27 | NaN | 2 | 591 | 1 | 2 | |

5 rows × 32 columns

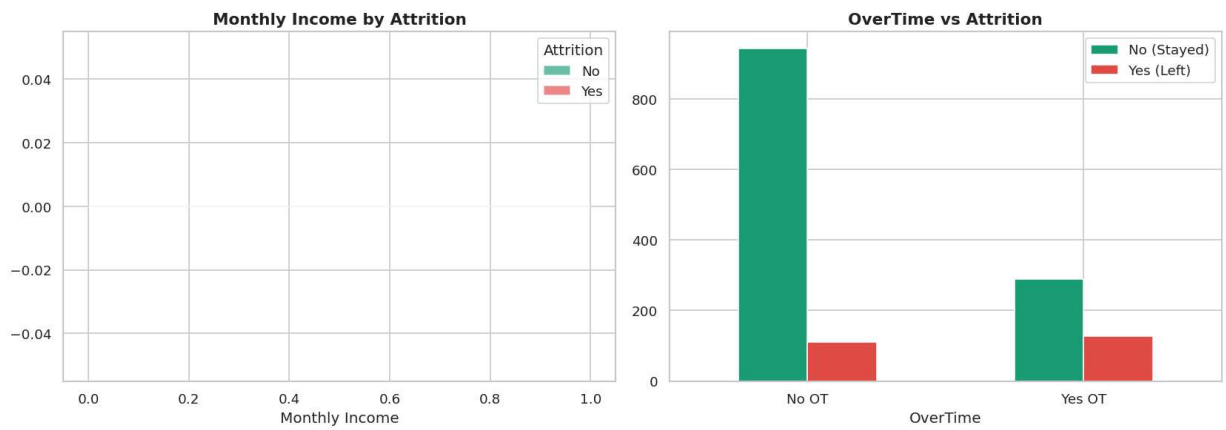
```
# FIXED - add errors='ignore'
df.drop(['EmployeeCount', 'Over18', 'StandardHours'],
        axis=1, inplace=True, errors='ignore')
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: Overall attrition count
counts = df_orig['Attrition'].value_counts()
axes[0].bar(counts.index, counts.values,
           color=['#1D9E75', '#E24B4A'], width=0.5, edgecolor='white')
for i, (k, v) in enumerate(counts.items()):
    axes[0].text(i, v+15, str(v), ha='center', fontsize=13, fontweight='bold')
axes[0].set_title('Employee Attrition Count', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Attrition')
axes[0].set_ylabel('Number of Employees')

# Plot 2: Department-wise attrition rate
dept = df_orig.groupby('Department')['Attrition'].value_counts(
    normalize=True).unstack() * 100
dept.plot(kind='bar', ax=axes[1], color=['#1D9E75', '#E24B4A'], edgecolor='whit
axes[1].set_title('Department-wise Attrition Rate (%)', fontsize=14, fontweight
axes[1].set_xlabel('Department')
axes[1].set_xticklabels(axes[1].get_xticklabels(), rotation=15)
axes[1].legend(['No (Stayed)', 'Yes (Left)'])

plt.tight_layout()
plt.show()
```

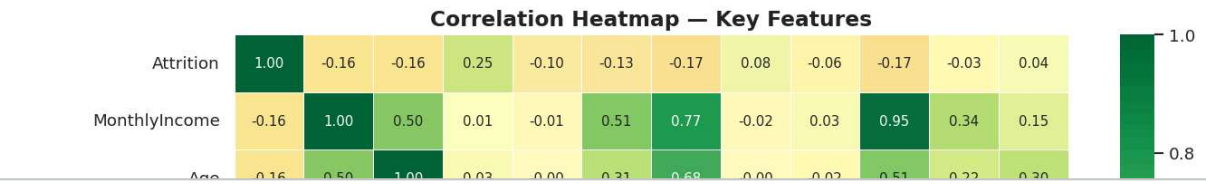

Employee Attrition Count
1233
237
Number of Employees
0
200
400
600
800
1000
1200
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Attrition


Department-wise Attrition Rate (%)
No (Stayed)
Yes (Left)
0
20
40
60
80
0
1
2
Department

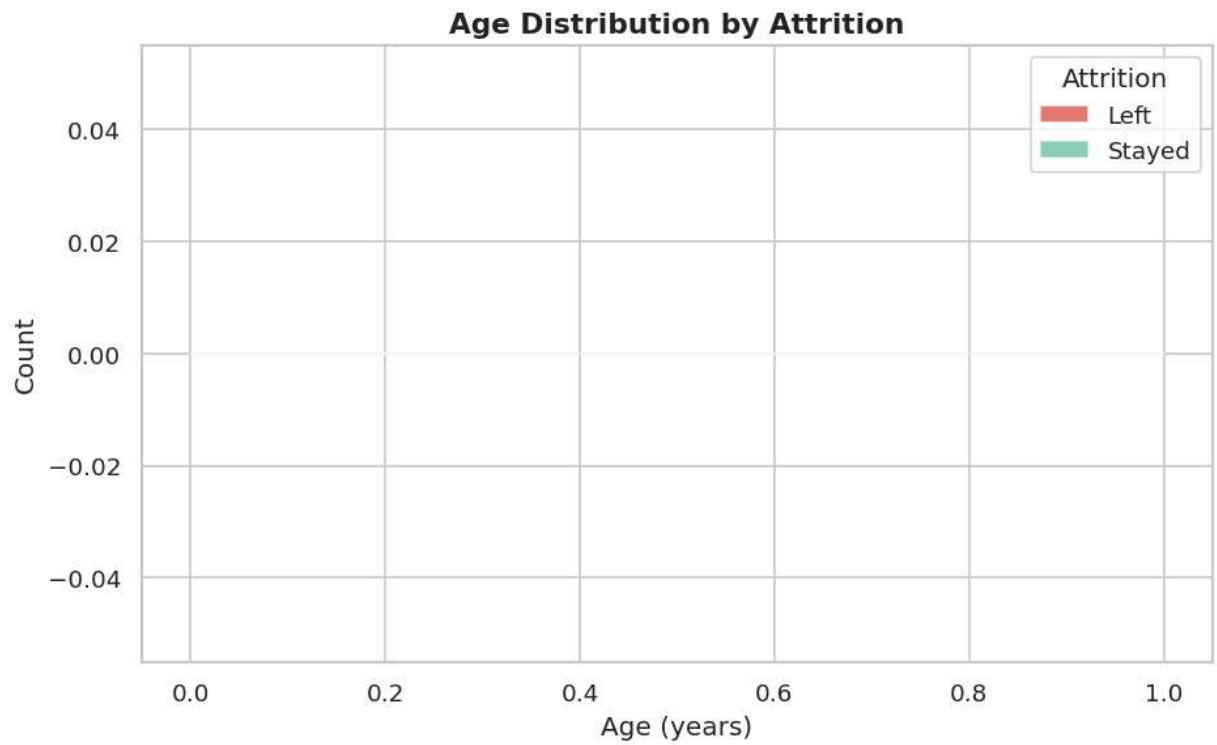
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 3: Monthly income by attrition
for label, color in [('No', '#1D9E75'), ('Yes', '#E24B4A')]:
    df_orig[df_orig['Attrition']==label]['MonthlyIncome'].hist(
        bins=30, ax=axes[0], alpha=0.65, color=color,
        label=label, edgecolor='white')
axes[0].set_title('Monthly Income by Attrition', fontsize=13, fontweight='bold'
axes[0].set_xlabel('Monthly Income')
axes[0].legend(title='Attrition')

# Plot 4: OverTime vs Attrition
ot = df_orig.groupby(['OverTime', 'Attrition']).size().unstack()
ot.plot(kind='bar', ax=axes[1], color=['#1D9E75', '#E24B4A'], edgecolor='white'
axes[1].set_title('OverTime vs Attrition', fontsize=13, fontweight='bold')
axes[1].set_xticklabels(['No OT', 'Yes OT'], rotation=0)
axes[1].legend(['No (Stayed)', 'Yes (Left)'])

plt.tight_layout()
plt.show()
```



```
top_cols = ['Attrition', 'MonthlyIncome', 'Age', 'OverTime',
            'JobSatisfaction', 'YearsAtCompany', 'TotalWorkingYears',
            'DistanceFromHome', 'WorkLifeBalance', 'JobLevel',
            'YearsSinceLastPromotion', 'NumCompaniesWorked']

plt.figure(figsize=(12, 9))
sns.heatmap(df[top_cols].corr(),
            annot=True, fmt='.2f',
            cmap='RdYlGn', center=0,
            linewidths=0.5,
            annot_kws={'size': 9})
plt.title('Correlation Heatmap — Key Features',
         fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(8, 5))
df_orig[df_orig['Attrition']=='Yes']['Age'].hist(
    bins=20, alpha=0.75, color='#E24B4A', label='Left', edgecolor='white')
df_orig[df_orig['Attrition']=='No']['Age'].hist(
    bins=20, alpha=0.5, color='#1D9E75', label='Stayed', edgecolor='white')
plt.title('Age Distribution by Attrition', fontsize=13, fontweight='bold')
plt.xlabel('Age (years)')
plt.ylabel('Count')
plt.legend(title='Attrition')
plt.tight_layout()
plt.show()
```

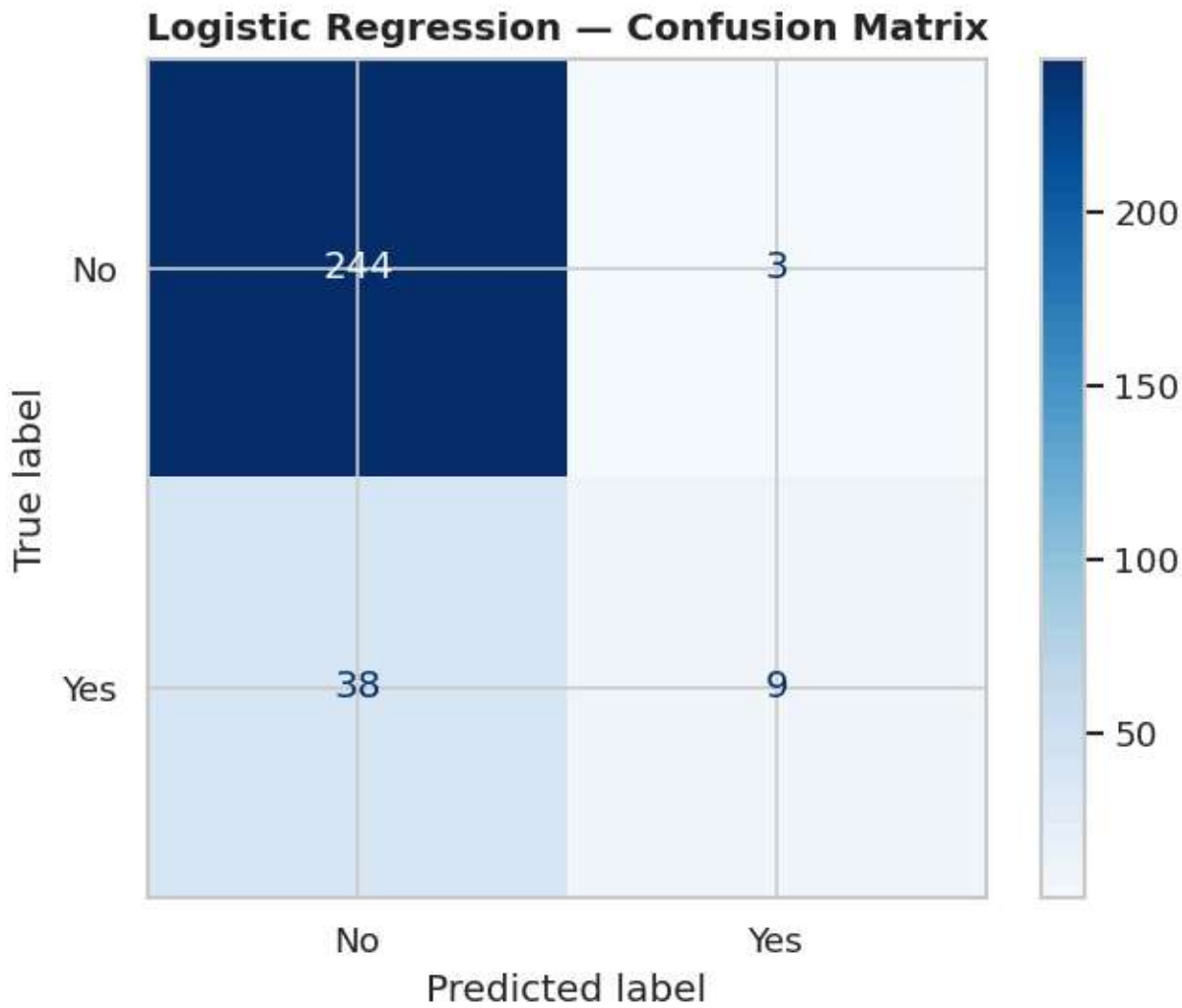


```
# Separate features (X) and target (y)
X = df.drop('Attrition', axis=1)
y = df['Attrition']

# 80% train, 20% test — stratify keeps class ratio balanced
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print(f"✅ Train size : {X_train.shape}")
print(f"   Test size  : {X_test.shape}")
print(f"   Attrition in train: {y_train.mean()*100:.1f}%")
print(f"   Attrition in test : {y_test.mean()*100:.1f}%")
```

```
✅ Train size : (1176, 31)
   Test size  : (294, 31)
   Attrition in train: 16.2%
   Attrition in test : 16.0%
```

```
# Train the model
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)

print("=== Logistic Regression Results ===")
print(f"Accuracy : {accuracy_score(y_test, y_pred_lr)*100:.1f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_lr,
      target_names=['No (Stayed)', 'Yes (Left)']))

# Confusion matrix
ConfusionMatrixDisplay.from_estimator(
    lr, X_test, y_test,
    display_labels=['No', 'Yes'],
    cmap='Blues')
plt.title('Logistic Regression — Confusion Matrix', fontweight='bold')
plt.show()
```

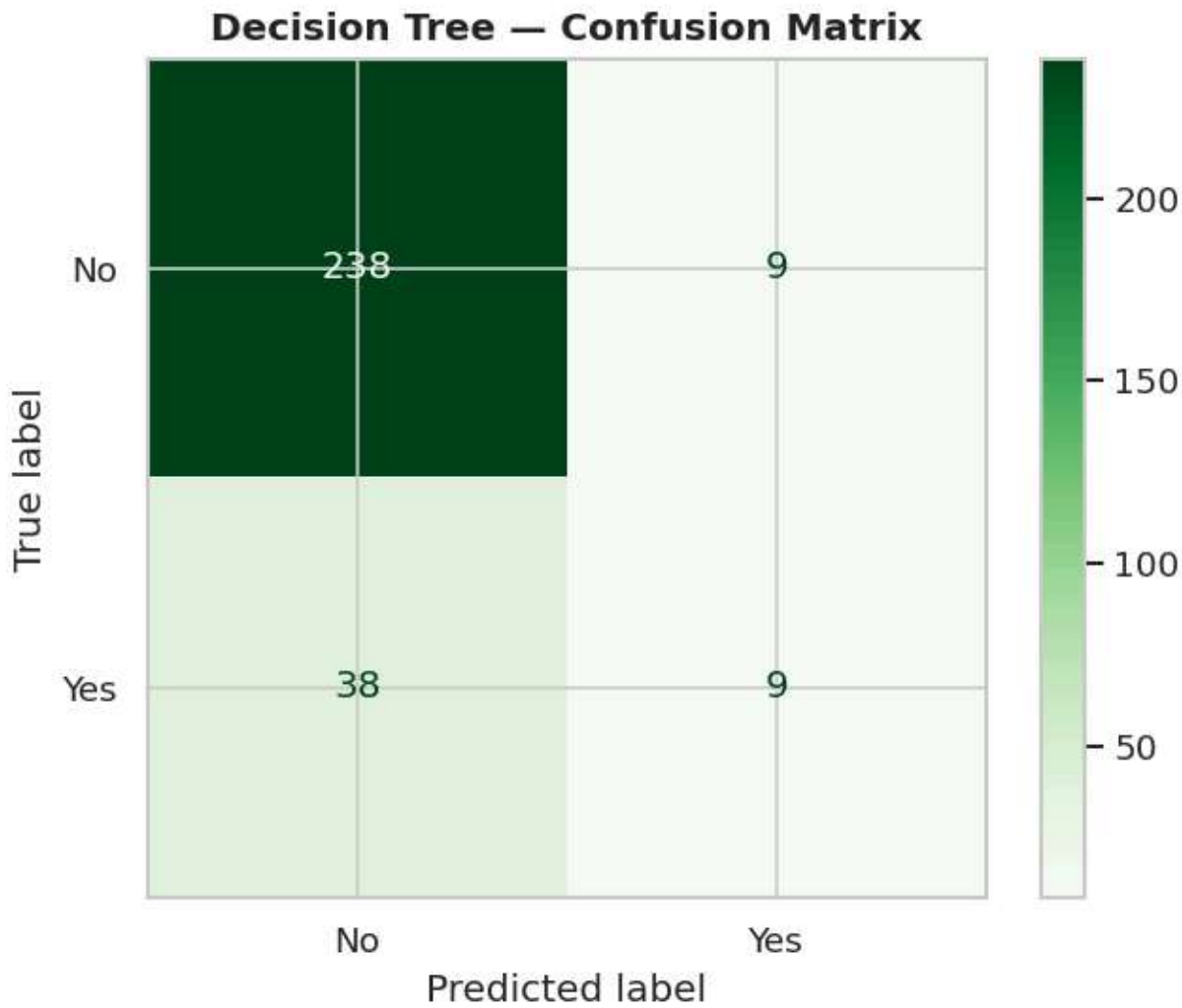
```
=== Logistic Regression Results ===
Accuracy : 86.1%

Classification Report:
              precision    recall  f1-score   support

 No (Stayed)       0.87      0.99      0.92       247
  Yes (Left)       0.75      0.19      0.31        47

    accuracy                           0.86       294
   macro avg       0.81      0.59      0.61       294
weighted avg       0.85      0.86      0.82       294
```




```
# Train Decision Tree
dt = DecisionTreeClassifier(max_depth=5, random_state=42)
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)

print("=== Decision Tree Results ===")
print(f"Accuracy : {accuracy_score(y_test, y_pred_dt)*100:.1f}%")
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred_dt,
      target_names=['No (Stayed)', 'Yes (Left)']))

# Confusion matrix
ConfusionMatrixDisplay.from_estimator(
    dt, X_test, y_test,
    display_labels=['No', 'Yes'],
    cmap='Greens')
plt.title('Decision Tree — Confusion Matrix', fontweight='bold')
plt.show()
```

```
=== Decision Tree Results ===
Accuracy : 84.0%

Classification Report:
              precision    recall  f1-score   support

 No (Stayed)       0.86      0.96      0.91       247
  Yes (Left)       0.50      0.19      0.28        47

    accuracy                           0.84       294
   macro avg       0.68      0.58      0.59       294
weighted avg       0.80      0.84      0.81       294
```




```
lr_acc = accuracy_score(y_test, y_pred_lr)
dt_acc = accuracy_score(y_test, y_pred_dt)

print("╔═══════════════════════════════════╗")
print("║      MODEL COMPARISON            ║")
print("╟═══════════════════════════════════╟")
print(f"║ Logistic Regression : {lr_acc*100:.1f}%      ║")
print(f"║ Decision Tree       : {dt_acc*100:.1f}%      ║")
```

```
print("╚═══════════════════════════════════╝")

best = "Logistic Regression" if lr_acc > dt_acc else "Decision Tree"
print(f"\n🏆 Best model: {best}")
```

```
╔═══════════════════════════════════╗
║      MODEL COMPARISON             ╕
╞═══════════════════════════════════╡
│ Logistic Regression : 86.1%       │
│ Decision Tree       : 84.0%       │
╘═══════════════════════════════════╛

🏆 Best model: Logistic Regression
```

```
# Keep original copy FIRST — before any changes
df_orig = df.copy()   # ← MOVE THIS TO THE TOP

# Drop useless columns
df.drop(['EmployeeCount', 'Over18', 'StandardHours'],
        axis=1, inplace=True, errors='ignore')

# Encode target column: Yes=1, No=0
df['Attrition'] = df['Attrition'].map({'Yes': 1, 'No': 0})

# Encode all categorical columns
le = LabelEncoder()
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = le.fit_transform(df[col])

print(f"✅ Cleaned! Shape: {df.shape}")
df.head()
```

```
✅ Cleaned! Shape: (1470, 32)
```

| | Age | Attrition | BusinessTravel | DailyRate | Department | DistanceFromHome | Educ |
|---|---|---|---|---|---|---|---|
| **0** | 41 | NaN | 2 | 1102 | 2 | 1 | |
| **1** | 49 | NaN | 1 | 279 | 1 | 8 | |
| **2** | 37 | NaN | 2 | 1373 | 1 | 2 | |
| **3** | 33 | NaN | 1 | 1392 | 1 | 3 | |
| **4** | 27 | NaN | 2 | 591 | 1 | 2 | |

5 rows × 32 columns

```
feat_imp = pd.Series(
    dt.feature_importances_,
    index=X.columns
```

```
).sort_values(ascending=False).head(15)

plt.figure(figsize=(10, 6))
colors_bar = ['#E24B4A' if i < 5 else '#534AB7'
              for i in range(len(feat_imp))]
feat_imp.plot(kind='bar', color=colors_bar, edgecolor='white')
plt.title('Top 15 Feature Importances (Decision Tree)',
         fontsize=13, fontweight='bold')
plt.xlabel('Feature')
plt.ylabel('Importance Score')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("\nTop 5 Attrition Drivers:")
for i, (feat, score) in enumerate(feat_imp.head(5).items(), 1):
    print(f"  {i}. {feat}: {score:.4f}")
```